

EDUCATION

Beijing Academy, Beijing, China

September 2023 – Present

Unweighted GPA: 4.0 SAT: 1560 TOEFL: 116

AP Scores (all 5):

- *Mathematics*: Precalculus, Calculus BC, Statistics
- *Science*: Physics C: Mechanics, Biology, Computer Science A
- *English and Social Science*: English Language & Composition, European History, Microeconomics, Macroeconomics
- *AP Capstone*: Seminar, Research

EXTRACURRICULAR ACTIVITIES

Orinav: navigation app for people with visual impairments, Founder & Lead Developer

October 2024 – Present

- Led team of 5 to refine app (22 months). Developed crossing assistance and obstacle & traffic light recognition algorithms.
- Reduced user navigation time by 12% and hesitations by 44%. Launched on iOS and Android. Scaled to 700+ users in 5 countries.
- Spoke at Inclusive Innovations 2025 and the 12th Beijing Academy Seminar; reported by *The Beijing News* and AbilityNet.

A11yLab: the Beijing Academy student accessibility initiative, Founder & Activity Coordinator

July 2024 – Present

- Hosted campus White Cane Experience for 122 community members and distributed 463 accessibility education leaflets.
- Launched inclusive group outings for 50+ participants; earned disability NGO Hequn (hequn.me)'s support.
- Contributed through design and execution to Tsinghua University project launching China's first youth accessibility awareness study.

An Introduction to Computer Science and the Art of Programming, Course Designer & Instructor

September 2023 – Present

- Designed 28-lesson curriculum covering basics often skipped in college courses (paths, commands, git). 95% would recommend.
- Teach 100+ students over 200+ hours for 3 years and ongoing: 60+ on campus, 44 by collaboration with Zhoushan Nanhai School.

The Whale Café: the Beijing Academy student-run café, CTO & Ordering Web App Developer

March 2024 – Present

- Fixed missing/unpaid orders caused by inefficient manual workflow: designed tailored quota, connected Weixin Pay, visualized data.
- Serves 40+ daily. Recovered ~30% lost revenue and raised payment rate from 55% to 98%. Saved ~CN¥10K in development.

Beijing Academy Website Modernization Project, Lead Developer

February 2024 – Present

- Revived twice-stalled project as its 3rd lead: automated content publishing with AI translation, standardized review procedures.
- Synchronized 400+ articles over past 5 years. Cut average update time by 79%. Improved search visibility by up to 30%.

AFS Global STEM Academies: a full scholarship sustainability program co-built with UPenn, Scholar

July 2025 – August 2025

- Earned full scholarship for 145 among 7000+ entrants. Studied sustainability in Egyptian villages with peers from 11 countries.
- Led 6-member multicultural team to build WasteMap. Mapped Cairo litter hotspots with heatmap visualization. Ranked #2/8 teams.

RESEARCH

- Lin, D., Jiang, B. B., Zhao, W., & Wang, Z. (2026). *Orinav: A hybrid mobile outdoor navigation application for people with visual impairments* [under peer review]. *Assistive Technology*.
- Lin, D. (2025). *Advancements in hardware accelerators for sequence alignment algorithms: A review*. — Mentor: Prof. Mark Bailey
- Lin, D., & Li, H. (2024). *Multinomial logistic regression with differentiable L1 regularization* — Mentor: Prof. Mark Girolami
- Lin, D., Li, T., Sun, Y., & Li, J. (2024). *To play or not to play: Modeling future Olympic Games*. High School Mathematical Contest in Modeling.
- Lin, D., Li, J., Liu, S., & Tang, Z. (2023). *Mathematical model of factors affecting pollutant concentrations in Lake Taihu*. International Mathematical Modeling Challenge.
- Lin, D., & Liu, B. (2026). *One event, three narratives: A cross-regional media case study of China's Japan travel advisory*.
- Lin, D. (2024). *Companion animal cloning: An ethical quandary*. AP Seminar Individual Research Report.
- Lin, D. (2024). *Whistleblowing: Advantages outweigh drawbacks*. AP Seminar Individual Written Argument.

AWARDS

Tech4Good Awards, Highly Commended

AbilityNet, 2026

- Orinav was among 3 of 36 finalists from 200+ entries awarded Highly Commended.

High School Mathematical Contest in Modeling, Finalist

COMAP, 2024

- Top 5% of teams awarded. Paper among 11 featured for outstanding model development out of 56 finalists.

Conrad Challenge China, Silver Medal

Space Center Houston, 2025

- Orinav was among the Top 13 of 200+ entries.

Chaoyang District Volunteer Service Competition, Winner

Chaoyang District Government, 2026

- 5 winners were selected from 168 participants. #1 among 44 high school students.

USA Computing Olympiad, Gold Division

USACO, 2025

Stanford e-China, Student Honoree

SPICE, 2026

- Only 46 students were admitted to the course, and 6 were selected for the award.

SKILLS AND INTERESTS

- *Development & Analysis*: Python, JavaScript, Java, C++, React, Flutter, iOS/Android, SQL, pandas, numpy
- *Research*: Needs assessment, user research, paper writing, data analysis, study design
- *Communication*: Project management, curriculum design, intercultural communication, public speaking
- *Interests*: CS instruction, musical performances, sight singing, choir, tennis
- *Language*: Mandarin Chinese (native), English (bilingual), French (limited working)